1958GOAM...24A...

SECTION A
MERIDIAN ASTRONOMY

Transit-Circle

Sun, Moon and Planets

Observed Right Ascensions, Declinations and Diameters

Compared with the Corresponding Results
of the Nautical Almanac, with Deduced Corrections
in Longitude and Latitude for the Sun and Moon

1948

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1958BOAM...24A...

SUN

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
1948 d h			h m s	s		° ′ ″	″	m s	s	′ ″	″
Jan. 9 12	GW		19 18 38.74	+.16		-22 13 2.69	-1.29	2 21.09	-.11	32 36.59	+1.73
15 12	GW	..	19 44 (40)	..		-21 16 10.92	-0.32	..	..	32 34.83	+0.49
19 12	GW		20 1 48.41	+.16		-20 30 1.07	+1.43	2 19.40	-.08	32 30.02	-3.80
20 12	FJ		20 6 3.53	+.12		-20 17 32.09	-0.49	2 19.20	-.08	32 30.56	-3.10
22 12	RL		20 14 31.54	+.11		-19 51 19.93	+1.77	2 18.83	-.03	32 35.06	+1.74
28 12	KB		20 39 36.64	+.13		-18 24 12.63	+0.37	2 17.49	-.05	32 31.78	-0.24
31 12	HB	1	20 51 58.35	+.10		-17 36 3.31	+1.19	..	..	32 30.77	-0.45
Feb. 4 12	FJ		21 8 16.13	+.12		-16 27 32.15	-0.45	2 16.03	+.09	32 28.91	-1.11
6 12	RL		21 16 20.24	+.17		-15 51 31.51	-1.01	2 15.47	-.01	32 31.03	+1.67
12 12	KB		21 40 13.24	+.11		-13 57 7.53	-0.33	2 14.06	-.08	32 27.93	+0.67
16 12	RL		21 55 53.03	+.11		-12 36 11.72	-1.72	2 13.27	-.01	32 25.62	-0.14
17 12	KB		21 59 46.05	+.06		-12 15 23.75	+0.75	2 12.94	-.14	32 28.01	+2.63
18 12	GW		22 3 38.37	+.05		-11 54 26.53	+0.87	2 12.49	-.39	32 22.40	-2.58
25 12	HB	2	22 30 25.43	-.02		- 9 22 58.14	-1.04	..	..	32 21.36	-0.62
26 12	RL		22 34 12.62	+.10		- 9 0 41.32	+0.38	2 11.44	+.04	32 25.47	+3.95
Mar. 1 12	HB		22 49 15.30	-.02		- 7 30 23.13	+0.17	2 10.73	-.03	32 17.36	-2.22
10 12	FJ	..	23 22 (40)	..		- 4 1 26.87	+0.03	..	..	32 16.46	+1.46
11 12	HB		23 26 20.52	-.07		- 3 37 53.21	+0.49	2 9.56	.00	32 15.95	+1.49
12 12	RL		23 30 1.21	+.11		- 3 14 15.88	+1.92	2 9.50	+.04	32 16.18	+2.24
13 12	KB		23 33 41.35	+.03		- 2 50 38.62	+1.18	2 9.38	.00	32 13.48	+0.06
14 12	GW	1	23 37 21.27	+.02		- 2 26 58.85	+1.15	..	..	32 11.93	-0.97
18 12	KB		23 51 58.57	+.03		- 0 52 11.65	-0.95	2 9.03	-.03	32 11.60	+0.80
22 12	RL		0 6 32.99	+.14		+ 0 42 35.25	-1.15	2 8.94	+.02	32 11.68	+3.02
23 12	KB	1	0 10 11.16	+.03		+ 1 6 15.14	+0.04	..	..	32 7.89	-0.23
24 12	GW		0 13 49.46	+.13		+ 1 29 51.71	-0.19	2 8.87	-.01	32 5.34	-2.22
Apr. 5 12	HB		0 57 30.47	.00		+ 6 8 48.00	-0.30	2 9.02	-.10	32 0.24	-0.58
6 12	KB		1 1 9.80	+.05		+ 6 31 30.72	+0.02	2 9.01	-.17	32 1.37	+1.11
7 12	RL		1 4 49.19	-.06		+ 6 54 8.86	+2.46	2 9.24	.00	32 0.46	+0.76
12 12	KB		1 23 10.53	+.08		+ 8 45 13.12	+0.32	2 9.57	-.07	31 55.97	-1.01
13 12	GW	..	1 26 (52)	..		+ 9 7 1.69	+0.59	..	..	31 53.06	-3.40
15 12	HB		1 34 14.66	+.01		+ 9 50 9.68	-0.12	2 10.03	+.09	31 54.28	-1.12
16 12	RL		1 37 56.73	+.02		+10 11 28.67	-0.83	2 10.20	+.16	31 57.64	+2.76
19 12	FJ		1 49 5.23	+.18		+11 14 26.53	+0.63	2 10.31	-.07	31 53.98	+0.64
20 12	HB		1 52 48.63	+.02		+11 35 3.90	+1.10	2 10.50	.00	31 51.95	-0.87
21 12	RL		1 56 32.65	+.07		+11 55 29.14	+1.04	2 10.73	+.09	31 54.06	+1.74
24 12	FJ	1	2 7 47.14	-.01		+12 55 32.06	-0.04	..	..	31 51.04	+0.26

Jan. 9. Observed through cloud; limbs barely visible.
15. Limbs very faint; observed through cloud.
20. Limbs boiling.
28. Clouds passing; limbs wavy.
Feb. 4. Limbs boiling; clouds passing for SL.
12. Limbs tremulous.
16. Limbs ill-defined.
17. Observed through cloud.
18. Very unsteady; clouds passing.
26. Limbs very unsteady.
Mar. 10. Limbs boiling.
Mar. 11. Limbs diffused and unsteady.
13. Diffused.
18. Clouds passing.
23. Clouds passing.
Apr. 5. Limbs unsteady.
12. Extremely diffused and unsteady.
15. Limbs diffused and unsteady.
16. Clouds passing; unsteady.
20. Limbs unsteady.
21. Unsteady and diffused.
24. Very ill-defined; clouds passing.

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SUN

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
1948 d h			h m s	s		° ′ ″	″	m s	s	′ ″	″
Apr. 26 12	KB		2 15 19.29	+.06		+13 34 30.15	−1.55	2 11.46	+.14	31 52.23	+2.45
27 12	RL	2	2 19 6.15	+.13		+13 53 41.75	+0.05	..	..	31 49.91	+0.63
28 12	GW		2 22 53.44	+.10		+14 12 38.89	+0.79	2 11.28	−.34	31 46.43	−2.35
30 12	HB		2 30 29.52	−.05		+14 49 46.46	−2.44	2 11.94	+.02	31 46.03	−1.75
May 6 12	RL		2 53 31.62	−.01		+16 35 21.46	−0.14	2 12.82	−.04	31 44.01	−0.91
7 12	KB		2 57 24.04	+.06		+16 52 2.93	+1.63	2 13.13	−.11	31 46.80	+2.34
8 12	GW	1	3 1 16.82	−.08		+17 8 23.68	−0.42	..	..	31 44.77	+0.77
13 12	GW		3 20 50.13	+.04		+18 25 55.93	+1.13	2 13.87	−.15	31 39.39	−2.51
18 12	GW		3 40 37.20	.00		+19 35 39.15	+0.75	2 14.75	−.07	31 37.96	−2.02
19 12	FJ		3 44 36.27	+.02		+19 48 34.52	−1.48	2 14.97	−.01	31 42.01	+2.39
21 12	RL		3 52 36.08	+.13	N	+20 13 27.47	−2.63	2 15.45	+.17	..	..
22 12	KB		3 56 36.62	+.03		+20 25 24.78	−1.42	2 15.47	+.03	31 39.38	+0.84
26 12	RL		4 12 44.40	+.06		+21 9 38.57	−0.13	2 15.86	−.14	31 38.71	+1.53
27 12	KB		4 16 47.59	+.05		+21 19 46.63	−1.17	2 16.02	−.12	31 37.95	+1.09
June 2 12	GW	1	4 41 16.67	+.19		+22 12 52.81	+0.31	..	..	31 32.39	−2.63
7 12	GW		5 1 51.65	+.02		+22 46 30.76	+0.46	2 17.05	−.27	31 31.24	−2.50
9 12	HB		5 10 7.87	.00		+22 57 10.10	−0.30	2 17.28	−.18	31 32.71	−0.59
10 12	RL		5 14 16.51	+.16		+23 1 51.92	−2.18	2 17.46	−.08	31 34.14	+1.04
11 12	KB		5 18 25.16	+.11		+23 6 11.39	−2.01	2 17.47	−.13	31 35.02	+2.12
15 12	RL		5 35 1.53	+.09		+23 19 24.95	−0.25	2 17.75	−.03	31 32.81	+0.59
16 12	KB		5 39 10.86	+.03		+23 21 40.78	−0.72	2 17.80	.00	31 32.09	+0.01
18 12	FJ	1	5 47 29.87	+.09	N	+23 24 58.40	−1.50	..	..	..	..
19 12	HB		5 51 39.31	.00		+23 26 0.74	−1.16	2 17.80	−.04	31 31.91	+0.23
July 5 12	GW		6 57 59.40	+.13		+22 46 42.62	−0.28	2 16.82	−.28	31 28.47	−2.17
14 12	KB		7 34 46.38	+.07		+21 39 40.31	−1.29	2 15.93	−.11	31 31.39	+0.21
19 12	GW		7 54 55.46	+.08		+20 49 23.88	−0.02	2 15.03	−.27	31 31.12	−0.68
20 12	FJ		7 58 55.50	−.05	N	+20 38 16.44	+0.04	2 15.24	+.08	..	..
22 12	KB		8 6 54.37	+.15		+20 14 58.64	−0.66	2 14.75	−.09	31 33.60	+1.36
26 12	HB		8 22 44.78	+.08		+19 24 22.37	−1.23	2 14.08	−.08	31 32.25	+0.67
27 12	KB		8 26 40.90	+.03		+19 10 54.89	−0.81	2 13.95	−.05	31 35.43	+2.31
28 12	GW		8 30 36.37	−.08		+18 57 7.95	−0.85	2 13.50	−.32	31 32.44	−0.86
29 12	RL		8 34 31.41	−.04		+18 43 3.17	−0.03	2 13.87	+.23	31 36.54	+3.02
30 12	FJ		8 38 25.92	+.06		+18 28 38.62	−0.38	2 13.50	+.02	31 34.62	+0.88
Aug. 14 12	KB		9 35 50.71	+.07		+14 18 54.04	−0.36	2 10.91	−.03	31 38.03	−0.05
18 12	KB		9 50 47.35	+.05		+13 2 58.50	−0.20	2 10.28	−.04	31 40.02	+0.46

Apr. 26. Extremely diffused and unsteady.
28. Very unsteady, through cloud.
30. Limbs very diffused and unsteady.
May 7. Extremely unsteady.
13. Through cloud and high fog.
18. Definition very bad.
19. Extremely unsteady.
22. Extremely diffused, unsteady, sky veiled.
26. Through cloud.
June 2. Observed through cloud.
7. Clouds passing.
9. Limbs very diffused and unsteady.
11. Limbs unsteady.
June 18. Clouds passing continually.
19. Diffused and very unsteady.
July 5. Clouds passing.
14. Clouds passing.
19. Definition extremely bad.
20. Very ill-defined, clouds passing.
22. Very unsteady, clouds passing.
26. Unsteady.
27. Very unsteady.
28. Definition very poor.
29. Limbs diffused.
Aug. 14. Through cloud.
18. Faint and unsteady.

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SUN

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1948 d h			h m s	s		° ′ ″	″	m s	s	′ ″	″
Aug. 22 12	KB		10 5 36.02	+.07		+11 43 44.80	-0.70	2 9.68	-.10	31 42.55	+1.47
24 12	RL		10 12 57.63	+.05	N	+11 3 1.76	+1.06	2 9.55	+.03	..	..
27 12	GW		10 23 57.08	+.07		+10 0 35.09	-0.01	2 8.93	-.23	31 40.88	-2.22
29 12	KB		10 31 14.90	+.09		+ 9 18 8.68	-0.52	2 8.93	-.01	31 44.36	+0.42
30 12	FJ		10 34 53.25	+.04		+ 8 56 43.53	+0.93	2 8.80	-.04	31 44.48	+0.10
31 12	GW	1	10 38 31.49	+.20		+ 8 35 6.34	-0.86	..	..	31 43.11	-1.71
Sept. 6 12	GW		11 0 14.00	+.03		+ 6 22 50.48	-1.22	2 8.00	-.28	31 45.06	-2.54
7 12	KB		11 3 50.32	+.09		+ 6 0 24.47	-0.73	2 8.08	-.14	31 48.41	+0.31
8 12	FJ		11 7 26.35	+.06		+ 5 37 51.53	-1.27	2 8.13	-.05	31 49.45	+0.85
13 12	KB		11 25 24.19	+.03		+ 3 43 53.50	-0.80	2 7.96	-.04	31 54.53	+3.37
16 12	KB		11 36 9.81	+.12		+ 2 34 40.65	-1.05	2 7.95	-.01	31 52.59	-0.13
17 12	FJ		11 39 44.96	+.16		+ 2 11 30.11	-1.09	2 7.79	-.17	31 53.95	+0.71
20 12	FJ		11 50 30.30	+.04		+ 1 1 44.06	-0.84	2 8.04	+.04	31 56.05	+1.23
21 12	HB		11 54 5.56	+.04		+ 0 38 23.69	-1.91	2 7.98	-.04	31 53.23	-2.13
22 12	KB		11 57 40.94	+.05		+ 0 15 4.07	-0.83	2 8.14	+.10	31 56.68	+0.80
24 12	HB		12 4 52.13	+.13		- 0 31 42.07	-2.57	2 8.09	-.03	31 56.39	-0.55
25 12	KB		12 8 27.93	+.14		- 0 55 2.63	-0.13	2 8.04	-.12	31 58.32	+0.86
28 12	KB		12 19 16.45	+.08		- 2 5 12.53	-1.63	2 8.31	-.03	32 0.18	+1.12
30 12	GW		12 26 30.07	+.10		- 2 51 53.72	-1.12	2 8.37	-.13	31 58.63	-1.53
Oct. 2 12	FJ		12 33 44.88	+.19		- 3 38 28.41	-0.91	2 8.69	+.03	32 1.56	+0.30
4 12	KB		12 41 0.70	+.02		- 4 24 54.32	-1.52	2 8.72	-.14	32 4.49	+2.13
5 12	FJ		12 44 39.45	+.27		- 4 48 1.90	-1.20	2 9.22	+.26	32 4.63	+1.71
9 12	GW	1	12 59 17.00	.00		- 6 19 51.80	+0.70	..	..	32 1.36	-3.82
10 12	KB		13 2 57.67	+.17		- 6 42 39.30	-0.80	2 9.62	+.02	32 6.03	+0.27
19 12	KB		13 36 24.90	+.02		-10 2 43.57	-2.17	2 10.97	-.09	32 12.02	+1.28
23 12	FJ		13 51 32.82	+.10		-11 28 4.78	-0.98	2 11.84	.00	32 12.88	0.00
26 12	FJ		14 3 1.10	+.11		-12 30 18.86	+0.24	2 12.48	+.04	32 15.37	+0.95
27 12	GW		14 6 52.01	+.10		-12 50 42.26	-0.46	2 12.40	-.26	32 11.83	-3.09
Nov. 17 12	GW	..	15 30 (47)	..		-19 2 6.68	-1.28	..	..	32 22.07	-2.89
20 12	GW		15 43 16.50	+.10		-19 44 19.84	-1.44	2 18.02	-.22	32 22.28	-3.88
24 12	FJ		16 0 7.01	+.07		-20 35 34.75	-0.55	2 19.11	-.01	32 27.14	-0.48
25 12	KB		16 4 21.66	+.08		-20 47 26.06	+0.84	2 19.27	-.05	32 30.47	+2.51
26 12	GW		16 8 37.11	+.11		-20 58 57.14	-0.74	2 19.13	-.39	32 27.04	-1.26
27 12	KB		16 12 53.13	-.04		-21 10 2.53	-0.23	2 19.73	+.01	32 29.44	+0.80

Aug. 22. Diffused and unsteady; thin cloud.
Sept. 7. SL through cloud.
13. Unsteady.
16. Through thin cloud; faint and diffused.
17. Extremely unsteady.
20. Observed through cloud.
24. Clouds passing throughout; limbs unsteady.
25. Unsteady.
28. Clouds passing for SL.
Oct. 4. Diffused and unsteady.
Oct. 5. Limbs boiling.
9. Definition very poor.
10. Very unsteady.
19. Very unsteady.
23. Limbs boiling.
26. Badly defined; clouds continually passing.
Nov. 24. Limbs extremely ill-defined.
25. Very unsteady.
27. Fog passing.

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SUN

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
1948 d h			h m s	s		° ′ ″	″	m s	s	′ ″	″
Dec. 1 12	FJ		16 30 5.13	+.13		-21 50 24.08	-0.18	2 20.46	.00	32 28.99	-0.95
6 12	FJ		16 51 49.11	+.12		-22 31 23.80	-1.10	2 21.30	+.04	32 30.29	-1.13
15 12	KB		17 31 25.69	+.13		-23 17 7.23	-0.03	2 22.17	-.05	32 34.04	+0.54
16 12	FJ		17 35 51.36	+.12		-23 19 55.72	-1.32	2 22.42	+.14	32 33.51	+0.17
17 12	GW		17 40 17.35	+.21		-23 22 11.32	+2.38	2 22.32	-.02	32 30.52	-3.34
23 12	GW		18 6 55.58	+.09		-23 26 17.47	+0.63	2 22.16	-.30	32 31.20	-3.40
29 12	GW		18 33 33.91	+.12		-23 13 25.14	+0.66	2 22.03	-.23	32 33.70	-1.28

Dec. 6. Limbs extremely faint and cloudy.
15. Limbs tremulous.
Dec. 16. Limbs boiling.
17. Definition poor.

MEAN MONTHLY CORRECTIONS TO NEWCOMB'S PLACE OF THE SUN AS GIVEN IN THE NAUTICAL ALMANAC

Mean Day of the Month	Observed - Computed R.A.	Observed - Computed Dec.	Observed - Computed Long.	Observed - Computed Lat.	Number of Observations R.A.	Number of Observations Dec.
1948	s	″	″	″		
Jan. 21	+.133	+0.38	+1.91	-0.03	5½	7
Feb. 15	+.095	-0.32	+1.20	-0.77	7½	8
Mar. 15	+.047	+0.27	+0.76	-0.03	8	10
Apr. 18	+.042	+0.11	+0.62	-0.12	13	15
May 17	+.036	-0.27	+0.44	-0.38	9½	9½
June 12	+.069	-0.78	+0.90	-0.84	8	8½
July 22	+.043	-0.58	+0.71	-0.44	10	9½
Aug. 24	+.072	-0.16	+1.05	+0.23	7½	7½
Sept. 18	+.082	-1.17	+1.59	-0.58	13	13
Oct. 14	+.115	-0.79	+1.88	-0.09	8½	9
Nov. 24	+.064	-0.57	+0.99	-0.38	5	6
Dec. 15	+.131	+0.15	+1.79	+0.24	7	7

When one limb only of the Sun has been observed, half-weight is given to the observation in forming the above monthly means.

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1958GOAM...24A...

MOON

Universal Time		Observer	Limb	Observed R.A. From Limb	Observed R.A. From Mösting A	O - C Limb	O - C Mösting A	Limb	Observed Dec. From Limb	Observed Dec. From Mösting A	O - C Limb	O - C Mösting A	Correction for Motion R.A.	Correction for Motion Dec.
1948 d	h			h m s	s	s	s		° ′ ″	″	″	″	s	″
Jan. 18	17	KB	1	1 13 27.56	..	-.13	..	S	+ 4 52 52.65	..	-2.15	..	.00	-.03
19	18	GW	1	2 0 58.85	..	+.05	..	..	+10 34 (21)	..	..	..	.00	..
28	2	RL	2	10 25 27.03	..	-.14	..	..	+15 7 (12)	..	..	..	-.01	..
29	3	KB	2	11 23 13.33	13.37	-.25	-.21	S	+ 8 46 14.29	15.85	-2.01	-0.45	-.01	+.06
30	4	GW	2	12 17 18.88	..	-.25	..	S	+ 2 6 8.76	..	-0.94	..	-.01	+.07
Feb. 1	5	HB	2	13 59 53.71	53.74	-.11	-.08	S	-10 33 61.20	58.79	-1.60	+0.81	.00	+.02
Feb. 16	17	RL	1	2 32 44.10	..	-.05	..	S	+14 20 32.71	..	-4.19	..	.00	-.01
17	18	KB	1	3 25 1.46	..	-.01	..	S	+19 15 8.76	..	-2.14	..	.00	.00
25	1	FJ	2	10 52 5.85	..	-.05	..	N	+12 13 31.95	..	-0.05	..	.00	+.01
26	2	HB	2	11 49 2.01	2.07	-.22	-.16	S	+ 5 28 33.22	35.72	-1.88	+0.62	-.01	+.05
28	3	KB	2	13 36 12.84	12.83	-.16	-.17	S	- 8 3 18.49	18.01	-1.99	-1.51	-.01	+.04
29	4	GW	2	14 28 50.00	..	-.24	..	S	-14 2 26.83	..	-2.03	..	-.01	+.06
Mar. 1	5	FJ	2	15 22 4.64	4.60	-.05	-.09	S	-19 6 55.94	54.32	-0.54	+1.08	.00	+.01
Mar. 18	18	KB	1	6 5 28.26	..	+.11	..	N	+27 9 11.92	..	-0.58	..	+.01	.00
20	20	FJ	1	8 17 7.27	7.19	-.12	-.20	N	+24 49 58.54	57.25	+1.24	-0.05	-.01	+.02
21	21	HB	1	9 21 4.93	4.91	.00	-.02	N	+20 53 33.81	34.36	-1.29	-0.74	.00	.00
22	22	RL	1	10 21 53.87	53.90	-.14	-.11	N	+15 29 19.90	17.55	+1.90	-0.45	-.01	+.03
23	23	KB	1	11 19 31.13	31.17	-.16	-.12	N	+ 9 5 3.54	2.72	-0.46	-1.28	-.01	+.04
25	0	GW	1	12 14 40.50	40.44	-.10	-.16	N	+ 2 10 59.34	59.19	+0.74	+0.59	.00	+.04
Apr. 15	17	HB	1	6 50 42.41	..	+.01	..	N	+27 22 14.35	..	-1.05	..	.00	.00
16	18	RL	1	7 55 35.88	..	-.07	..	N	+25 52 49.73	..	-0.97	..	.00	+.01
18	20	GW	1	9 58 44.19	..	-.08	..	N	+17 50 9.84	..	-2.16	..	.00	+.02
19	21	FJ	1	10 55 37.83	37.83	-.14	-.14	N	+11 57 7.50	8.09	-1.00	-0.41	-.01	+.04
20	22	HB	1	11 50 1.85	1.92	-.09	-.02	N	+ 5 22 47.72	46.41	-0.78	-2.09	.00	+.02
22	24	KB	1	13 35 44.01	44.00	-.16	-.17	N	- 8 8 40.71	41.96	+0.79	-0.46	-.01	+.04
25	1	FJ	2	15 24 19.07	18.78	+.09	-.20	S	-19 33 50.35	49.63	-0.75	-0.03	.00	+.01
May 18	21	GW	1	12 23 33.10	..	+.04	..	N	+ 1 4 27.17	..	-0.13	..	.00	-.01
19	21	FJ	1	13 14 51.46	51.46	-.07	-.07	N	- 5 32 30.44	29.28	+1.76	+2.92	.00	+.02
21	23	RL	1	15 0 10.10	10.34	-.26	-.02	N	-17 23 56.31	57.78	+3.29	+1.82	-.01	+.03
22	24	KB	1	15 55 41.91	..	-.20	..	S	-21 58 50.67	..	-0.07	..	-.01	+.03
25	2	FJ	2	17 51 53.90	..	-.16	..	S	-27 5 35.26	..	-1.46	..	-.01	+.01
27	3	RL	2	19 46 59.49	..	+.03	..	N	-26 10 23.36	..	-0.16	..	.00	.00
28	4	KB	..	20 40 (53)	53.05	..	-.24	N	-23 41 33.73	32.32	-0.63	+0.78	-.01	-.03

Jan.30. SL mountainous.
Feb.25. Through cloud.
26. Observed through cloud.
Mar.20. Definition good.
21. Clouds passing.
Apr.19. Through thin cloud.
Apr.22. Diffused and unsteady.
25. Extremely ill-defined and unsteady.
May 22. Very indistinct; through thin cloud.
25. Through cloud.
28. Very faint.

Mösting A

Jan.29. Poor detail.
Feb.28. Indistinct.
May 28. No detail visible.

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MOON

Universal Time	Observer	Limb	Observed R.A. From Limb	Observed R.A. From Mösting A	O - C Limb	O - C Mösting A	Limb	Observed Dec. From Limb	Observed Dec. From Mösting A	O - C Limb	O - C Mösting A	Correction for Motion R.A.	Correction for Motion Dec.
1948 d h			h m s	s	s	s		° ′ ″	″	″	″	s	″
June 16 20	KB	1	13 49 26.52	26.53	-.05	-.04	N	- 9 53 13.10	14.67	0.00	-1.57	.00	+.01
18 22	FJ	1	15 35 4.08	4.07	-.21	-.22	N	-20 28 19.66	16.99	-0.46	+2.21	-.01	+.04
24 2	FJ	2	20 19 44.99	..	-.14	..	N	-24 42 46.73	..	-2.53	..	-.01	-.02
July 22 1	HB	2	20 53 23.89	..	-.12	..	S	-22 42 10.28	..	-1.18	..	.00	-.02
23 2	KB	2	21 43 23.92	23.98	-.13	-.07	N	-18 52 39.52	38.84	+0.18	+0.86	.00	-.02
26 4	FJ	..	23 59 (28)	28.23	..	-.10	N	- 3 41 31.57	31.69	-2.27	-2.39	.00	-.02
27 4	HB	2	0 43 5.13	5.36	-.33	-.10	N	+ 1 57 52.73	53.32	-0.87	-0.28	-.01	-.05
28 5	KB	2	1 27 40.08	40.16	-.21	-.13	N	+ 7 38 2.08	2.14	-2.32	-2.26	-.01	-.04
Aug. 14 20	KB	1	17 49 24.63	24.72	-.22	-.13	S	-27 14 3.77	2.63	-2.47	-1.33	-.01	+.01
17 23	HB	1	20 36 24.79	24.93	-.17	-.03	S	-23 36 36.90	35.46	-2.40	-0.96	.00	-.01
18 24	KB	1	21 27 10.24	10.18	-.13	-.19	S	-20 13 3.83	4.29	-0.13	-0.59	-.01	-.03
20 0	GW	2	22 15 6.76	6.82	-.19	-.13	S	-15 49 9.85	8.22	-1.85	-0.22	-.01	-.03
21 1	RL	2	23 0 44.42	..	-.17	..	N	-10 48 20.64	..	-0.54	..	-.01	-.04
23 2	KB	2	0 28 17.02	17.10	-.20	-.12	N	+ 0 15 29.62	28.52	-1.28	-2.38	-.01	-.04
24 3	GW	2	1 12 9.44	..	-.36	..	N	+ 5 56 33.59	..	-1.61	..	-.01	-.08
Sept. 11 19	FJ	1	18 28 17.67	..	-.06	..	S	-27 45 8.35	..	-3.25	..	.00	.00
14 22	FJ	1	21 11 9.28	9.11	+.22	+.05	S	-21 34 11.10	11.06	+0.60	+0.64	.00	+.02
16 23	KB	1	22 46 2.86	2.86	-.10	-.10	S	-12 30 53.27	52.64	-2.17	-1.54	.00	-.02
17 24	FJ	1	23 30 33.64	..	.00	..	S	- 7 9 18.82	..	-0.42	..	.00	.00
19 0	GW	2	0 14 13.53	..	-.18	..	S	- 1 30 18.01	..	-0.91	..	-.01	-.04
21 2	FJ	2	1 42 56.02	..	-.31	..	N	+ 9 54 20.75	..	-3.05	..	-.01	-.07
25 5	HB	2	5 13 18.55	18.50	+.01	-.04	N	+26 40 2.59	2.98	-2.11	-1.72	.00	.00
26 6	KB	2	6 15 40.34	..	-.02	..	S	+27 49 59.28	..	-0.02	..	.00	.00
Oct. 9 18	GW	1	19 4 27.61	..	+.01	..	S	-27 34 12.69	..	-1.59	..	.00	.00
13 21	FJ	1	22 30 12.91	12.93	-.15	-.13	S	-14 21 32.15	31.51	+0.25	+0.89	.00	-.03
14 22	KB	1	23 15 10.90	10.97	-.14	-.07	S	- 9 7 39.24	40.26	-1.14	-2.16	.00	-.02
15 22	GW	1	23 59 4.46	..	+.01	..	S	- 3 31 33.80	..	-0.20	..	.00	.00
19 0	GW	2	2 14 21.65	..	-.25	..	N	+13 35 46.74	..	-0.66	..	-.01	-.05
20 1	KB	2	3 4 4.48	4.55	-.22	-.15	N	+18 39 42.96	42.60	-0.34	-0.70	-.01	-.04
23 4	KB	2	5 56 12.06	12.03	-.05	-.08	N	+27 44 18.66	18.42	-1.54	-1.78	.00	.00
24 5	FJ	2	6 59 25.79	25.74	-.09	-.14	S	+27 40 61.31	59.95	+1.81	+0.45	-.01	.00
26 7	KB	2	9 4 24.14	..	-.11	..	S	+22 12 57.40	..	+0.40	..	.00	+.02
Nov. 25 7	FJ	2	11 34 41.09	..	-.07	..	S	+ 6 38 4.79	..	-0.11	..	.00	+.02
Dec. 7 18	KB	..	22 39 (3)	..	..	..	S	-13 14 59.88	..	-1.38	..	..	-.02
14 23	GW	..	4 9 (36)	..	..	..	S	+23 42 11.26	..	-2.14	..	..	-.02
15 24	KB	1	5 10 6.63	6.68	-.16	-.11	N	+26 34 45.75	45.07	-1.25	-1.93	-.01	-.01
23 6	FJ	2	12 10 52.26	52.15	-.05	-.16	S	+ 1 43 12.31	14.05	-0.89	+0.85	.00	+.03

June 16. Clouds passing at times. NL poor.
18. Clouds passing.
24. Observed through cloud.
July 26. Clouds passing.
Sept. 11. Limbs slightly tremulous.
14. Very unsteady; limbs tremulous.
21. Observed through cloud.

Oct. 13. Through cloud.
19. Definition extremely bad throughout.
24. Definition good.
Nov. 25. SL mountainous.
Dec. 7. Through cloud, very poor object.
15. Tremulous.

Mösting A. - Oct. 24. Definition good.

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1958GDAM...24A...

MOON

DEDUCED CORRECTIONS IN LONGITUDE AND LATITUDE TO BROWN'S "TABLES OF THE MOON".

The observed corrections to the tabular R.A. and Dec. of the Nautical Almanac are shown in the preceding pages. In R.A. they give a mean correction to the tables, obtained from complete lunations ending in the year, of $-^{s}.11$. This corresponds approximately to $-1''.6$ in mean longitude.

In order to allow for small periodic changes due to elliptic and other inequalities in the Moon's motion introduced by this correction to the mean longitude, corrections to the computed right ascension and declination have been deduced by retarding the Moon's predicted position $1''.6$ in mean longitude and making the corresponding corrections to the N.A. positions in R.A. and Dec. of $-.049\ \Delta\alpha$ and $-.049\ \Delta\delta$.

$\Delta\alpha$ and $\Delta\delta$ are the variations of the Moon's R.A. and Dec. in 1 minute of time.

The following residuals therefore represent the corrections to Brown's tables in R.A., Dec., deduced longitude and latitude after a correction of $-1''.6$ has been made to the tabular mean longitude.

The time of the new moon (●) is Universal Time.

RESIDUALS (O - C) AFTER CORRECTING BROWN'S TABLES FOR A DECREASE IN MEAN LONGITUDE OF $1''.6$

Universal Time			Lunar Day	Observed - Computed							
				R.A.		Declination		Longitude		Latitude	
1947-48				Limb	Mösting A	Limb	Mösting A	Limb	Mösting A	Limb	Mösting A
d	h	m		s	s	″	″	″	″	″	″
Dec. 12	12	53	●								
Dec. 30	2	22	69620	-.01	+.03	+0.6	-0.4	-0.3	+0.5	+0.5	-0.3
Jan. 11	7	44	●								
Jan. 18	17	25	69639	-.04		-1.4		-1.1		-1.1	
29	2	54	649	-.13	-.09	-2.9	-1.3	-0.6	-0.7	-3.4	-1.7
30	3	44	650	-.13		-1.8		-1.1		-2.4	
Feb. 1	5	18	652	-.01	+.02	-2.3	+0.1	+0.7	+0.2	-2.2	+0.2
Feb. 10	3	2	●								
Feb. 16	16	50	69667	+.05		-3.6		-0.4		-3.6	
17	17	39	668	+.10		-1.6		+1.0		-1.9	
25	0	37	675	+.07		-0.8		+1.3		-0.4	
26	1	30	676	-.10	-.04	-2.8	-0.3	-0.3	-0.4	-3.1	-0.5
28	3	9	678	-.04	-.05	-2.8	-2.3	+0.5	+0.2	-2.8	-2.4
29	3	57	679	-.12		-2.8		-0.8		-3.2	
Mar. 1	4	46	680	+.06	+.02	-1.1	+0.5	+1.1	+0.1	-0.8	+0.6
Mar. 10	21	15	●								
Mar. 18	18	21	69697	+.23		-0.5		+3.1		-0.5	
20	20	24	699	+.02	-.06	+0.8	-0.4	+0.1	-0.7	+0.9	-0.6
21	21	24	700	+.13	+.11	-1.9	-1.3	+2.3	+1.9	-1.2	-0.8
22	22	21	701	-.01	+.02	+1.2	-1.2	-0.6	+0.7	+1.0	-1.0
23	23	14	702	-.04	.00	-1.3	-2.1	0.0	+0.8	-1.4	-2.0
25	0	5	703	+.01	-.05	-0.2	-0.3	+0.2	-0.6	-0.1	-0.6
Apr. 9	13	16	●								
Apr. 15	17	16	69724	+.14		-1.1		+2.0		-1.0	
16	18	16	725	+.06		-1.3		+1.0		-1.1	
18	20	11	727	+.04		-2.8		+1.5		-2.4	

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1958GOAM...24A....

MOON

RESIDUALS (O - C) AFTER CORRECTING BROWN'S TABLES FOR A DECREASE IN MEAN LONGITUDE OF 1".6

Universal Time			Lunar Day	Observed - Computed							
				R.A.		Declination		Longitude		Latitude	
1948				Limb	Mösting A	Limb	Mösting A	Limb	Mösting A	Limb	Mösting A
d	h	m		s	s	"	"	"	"	"	"
Apr. 19	21	4	69728	-.02	-.02	-1.8	-1.2	+0.4	+0.2	-1.8	-1.2
20	21	54	729	+.02	+.09	-1.6	-2.9	+0.9	+2.4	-1.4	-2.2
22	23	32	731	-.04	-.05	0.0	-1.3	-0.5	-0.2	-0.3	-1.5
25	1	12	733	+.20	-.09	-1.3	-0.6	+3.1	-1.1	-0.6	-0.9
May 9	2	30	●								
May 18	20	38	69756	+.14		-0.9		+2.3		0.0	
19	21	25	757	+.03	+.03	+1.0	+2.1	0.0	-0.4	+1.0	+2.1
21	23	2	759	-.14	+.10	+2.6	+1.2	-2.7	+1.0	+2.0	+1.5
22	23	54	760	-.08		-0.6		-1.0		-0.8	
25	1	42	762	-.03		-1.6		-0.4		-1.6	
27	3	29	764	+.14		+0.1		+1.9		-0.3	
28	4	18	765		-.12		+1.2		-1.3		+1.6
June 7	12	55	●								
June 16	20	10	69784	+.05	+.06	-0.7	-2.3	+1.0	+1.6	-0.4	-1.8
18	21	47	786	-.09	-.10	-1.0	+1.6	-1.0	-1.8	-1.3	+1.3
24	2	11	791	-.02		-2.2		-0.8		-2.1	
July 6	21	9	●								
July 22	0	55	69818	-.02		-0.8		-0.5		-0.6	
23	1	41	819	-.03	+.03	+0.7	+1.4	-0.2	+0.9	+0.8	+1.2
26	3	44	822		-.01		-1.7		-0.8		-1.5
27	4	24	823	-.23	.00	-0.1	+0.5	-3.2	+0.2	+1.2	+0.4
28	5	5	824	-.11	-.03	-1.6	-1.5	-2.1	-1.0	-0.9	-1.3
Aug. 5	4	13	●								
Aug. 14	20	17	69841	-.09	.00	-2.6	-1.5	-1.2	0.0	-2.6	-1.5
17	22	52	844	-.06	+.08	-2.0	-0.6	-1.3	+0.9	-1.8	-0.8
18	23	38	845	-.02	-.08	+0.4	-0.1	-0.2	-1.1	+0.4	+0.3
20	0	22	846	-.09	-.03	-1.2	+0.4	-1.7	-0.3	-0.7	+0.5
21	1	4	847	-.07		+0.1		-0.9		+0.5	
23	2	23	849	-.10	-.02	-0.6	-1.6	-1.6	-0.9	+0.1	-1.4
24	3	3	850	-.26		-0.8		-3.9		+0.7	
Sept. 3	11	21	●								
Sept. 11	19	6	69868	+.06		-3.2		+0.6		-3.3	
14	21	36	871	+.32	+.15	+1.0	+1.1	+4.6	+2.3	-0.3	+0.4
16	23	3	873	-.01	-.01	-1.5	-0.9	-0.7	-0.5	-1.4	-0.8
17	23	43	874	+.09		+0.2		+1.3		-0.3	
19	0	23	875	-.08		-0.2		-1.2		+0.3	
21	1	44	877	-.21		-2.3		-3.7		-1.0	
25	4	58	881	+.13	+.08	-1.9	-1.5	+1.6	+1.0	-2.0	-1.6
26	5	56	882	+.11		0.0		+1.5		0.0	
Oct. 2	19	42	●								
Oct. 9	17	52	69895	+.13		-1.5		+1.6		-1.6	
13	21	1	899	-.06	-.04	+0.9	+1.5	-0.5	0.0	+1.1	+1.6
14	21	42	900	-.05	+.02	-0.5	-1.5	-0.9	-0.3	-0.1	-1.5
15	22	22	901	+.10		+0.5		+1.6		-0.2	

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1958GOAM...24A...

MOON

RESIDUALS (O - C) AFTER CORRECTING BROWN'S TABLES FOR A DECREASE IN MEAN LONGITUDE OF 1".6

Universal Time	Lunar Day	Observed - Computed							
		R.A.		Declination		Longitude		Latitude	
1948		Limb	Mösting A	Limb	Mösting A	Limb	Mösting A	Limb	Mösting A
d h m		s	s	"	"	"	"	"	"
Oct. 19 0 25	69904	-.14		0.0		-1.9		+0.7	
20 1 10	905	-.11	-.04	+0.3	-0.1	-1.4	-0.6	+0.7	+0.1
23 3 50	908	+.08	+.05	-1.4	-1.7	+1.0	+0.7	-1.4	-1.7
24 4 49	909	+.05	.00	+1.7	+0.3	+0.5	0.0	+1.8	+0.3
26 6 46	911	+.01		-0.2		+0.2		-0.1	
Nov. 1 6 2	●								
Nov. 25 7 18	69940	+.04		-0.9		+0.9		-0.6	
Nov. 30 18 44	●								
Dec. 15 23 32	69960	-.02	+.03	-1.0	-1.7	-0.4	+0.2	-1.0	-1.7
23 6 4	967	+.05	-.06	-1.7	0.0	+1.4	-0.8	-1.3	-0.4
Dec. 30 9 44	●								

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1958GOAM...24A...

MOON

CORRECTIONS TO THE MOON'S CENTRE IN R.A., DEC., LONGITUDE AND LATITUDE FROM OBSERVATIONS OF THE LIMB AND THE CRATER MÖSTING A AFTER A CORRECTION OF −1″.6 HAS BEEN MADE TO THE TABULAR MEAN LONGITUDE

(Means for each complete lunation ending in the year 1948)

Extent of Lunation	Number of Observations		Observed − Computed							
	Limb	Mösting A	R.A.		Declination		Longitude		Latitude	
1947–48			Limb	Mösting A	Limb	Mösting A	Limb	Mösting A	Limb	Mösting A
			s	s	″	″	″	″	″	″
Dec. 30	1	1	−.01	+.03	+0.6	−0.4	−0.3	+0.5	+0.5	−0.3
Jan. 18 to Feb. 1	4	2	−.08	−.04	−2.1	−0.6	−0.5	−0.2	−2.3	−0.8
Feb. 16 to Mar. 1	7	3	.00	−.02	−2.2	−0.7	+0.3	0.0	−2.3	−0.8
Mar. 18 to Mar. 25	6	5	+.06	.00	−0.3	−1.1	+0.8	+0.4	−0.2	−1.0
Apr. 15 to Apr. 25	7	4	+.06	−.02	−1.4	−1.5	+1.2	+0.3	−1.2	−1.4
May 18 to May 28	6	3	+.01	.00	+0.1	+1.5	0.0	−0.2	+0.1	+1.7
June 16 to June 24	3	2	−.02	−.02	−1.3	−0.3	−0.3	0.0	−1.3	−0.3
July 22 to July 28	4	4	−.10	.00	−0.4	−0.3	−1.5	−0.2	+0.1	−0.3
Aug. 14 to Aug. 24	7	5	−.10	−.01	−1.0	−0.7	−1.5	−0.3	−0.5	−0.6
Sept.11 to Sept.26	8	3	+.05	+.07	−1.0	−0.4	+0.5	+0.9	−1.0	−0.6
Oct. 9 to Oct. 26	9	5	.00	.00	0.0	−0.3	0.0	0.0	+0.1	−0.2
Nov. 25	1	−	+.04		−0.9		+0.9		−0.6	
Dec. 15 to Dec. 23	2	2	+.02	−.02	−1.4	−0.8	+0.5	−0.3	−1.1	−1.0

MEANS FOR THE YEAR 1948

	No. of Obs.	R.A.	Declination	Longitude	Latitude
		s	″	″	″
Limb	65	−.002	−0.90	+0.03	−0.78
Mösting A	39	−.002	−0.53	+0.06	−0.50

The above mean corrections to Brown's tables have been derived after correcting the tabular positions in R.A. and Dec. by retarding the Moon's predicted position 1″.6 in mean longitude.

The mean correction therefore to the tables for the year 1948 is:-

	Longitude	Latitude
	″	″
Limb	−1.57	−0.78
Mösting A	−1.54	−0.50

Note:- If Newcomb's equinox is adopted, the mean correction to Brown's tables in longitude becomes:-

	″
Limb	−0.90
Mösting A	−0.87

The corrections to Brown's tables given in the Greenwich Observations from 1923 to 1930 are based on Newcomb's equinox.

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1958GOAM...24A...

MOON

DEDUCTION OF A CORRECTION TO THE TABULAR HORIZONTAL DIAMETER FROM A COMPARISON OF THE OBSERVATIONS OF THE FIRST AND SECOND LIMB WITH THOSE OF THE CRATER MÖSTING A

Date	Observer	Observed *minus* Computed			Date	Observer	Observed *minus* Computed		
		Mösting A	☽ 1L	Mösting A *minus* ☽ 1L			☽ 2L	Mösting A	☽ 2L *minus* Mösting A
1948		s	s	s	1948		s	s	s
Mar. 20	FJ	−.20	−.12	−.08	Jan. 29	KB	−.25	−.21	−.04
21	HB	−.02	.00	−.02	Feb. 1	HB	−.11	−.08	−.03
22	RL	−.11	−.14	+.03	26	HB	−.22	−.16	−.06
23	KB	−.12	−.16	+.04	28	KB	−.16	−.17	+.01
25	GW	−.16	−.10	−.06	Mar. 1	FJ	−.05	−.09	+.04
Apr. 19	FJ	−.14	−.14	.00	Apr. 25	FJ	+.09	−.20	+.29
20	HB	−.02	−.09	+.07	July 23	KB	−.13	−.07	−.06
22	KB	−.17	−.16	−.01	27	HB	−.33	−.10	−.23
May 19	FJ	−.07	−.07	.00	28	KB	−.21	−.13	−.08
21	RL	−.02	−.26	+.24	Aug. 20	GW	−.19	−.13	−.06
June 16	KB	−.04	−.05	+.01	23	KB	−.20	−.12	−.08
18	FJ	−.22	−.21	−.01	Sept. 25	HB	+.01	−.04	+.05
Aug. 14	KB	−.13	−.22	+.09	Oct. 20	KB	−.22	−.15	−.07
17	HB	−.03	−.17	+.14	23	KB	−.05	−.08	+.03
18	KB	−.19	−.13	−.06	24	FJ	−.09	−.14	+.05
Sept. 14	FJ	+.05	+.22	−.17	Dec. 23	FJ	−.05	−.16	+.11
16	KB	−.10	−.10	.00					
Oct. 13	FJ	−.13	−.15	+.02					
14	KB	−.07	−.14	+.07					
Dec. 15	KB	−.11	−.16	+.05					
			Mean	+.018				Mean	−.008

Assuming that the effect of any error in the adopted Physical Libration of the Moon will disappear in the mean of these quantities, and allowing for the fact that the Tabular Diameter has been increased by $+^{s}.160$, the correction to the Moon's Tabular Horizontal Diameter is

$$+^{s}.160 \quad +^{s}.018 \quad -^{s}.008 \quad = \quad +^{s}.170$$

This result is independent of any error in the assumed selenographical position of the crater. The observations indicate that the Tabular R.A. of the crater, relatively to the Moon's centre, requires to be increased by

$$\tfrac{1}{2}\ (+^{s}.018 \quad +^{s}.008) \quad = \quad +^{s}.013$$

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MOON

DEDUCTION OF A CORRECTION TO THE TABULAR VERTICAL DIAMETER FROM A COMPARISON OF THE OBSERVATIONS OF THE NORTH AND SOUTH LIMB WITH THOSE OF THE CRATER MÖSTING A

Date	Observer	Observed *minus* Computed			Date	Observer	Observed *minus* Computed		
		Mösting A	☽ NL	☽ NL *minus* Mösting A			☽ SL	Mösting A	Mösting A *minus* ☽ SL
1948		″	″	″	1948		″	″	″
Mar. 20	FJ	−0.05	+1.24	+1.29	Jan. 29	KB	−2.01	−0.45	+1.56
21	HB	−0.74	−1.29	−0.55	Feb. 1	HB	−1.60	+0.81	+2.41
22	RL	−0.45	+1.90	+2.35	26	HB	−1.88	+0.62	+2.50
23	KB	−1.28	−0.46	+0.82	28	KB	−1.99	−1.51	+0.48
25	GW	+0.59	+0.74	+0.15	Mar. 1	FJ	−0.54	+1.08	+1.62
Apr. 19	FJ	−0.41	−1.00	−0.59	Apr. 25	FJ	−0.75	−0.03	+0.72
20	HB	−2.09	−0.78	+1.31	Aug. 14	KB	−2.47	−1.33	+1.14
22	KB	−0.46	+0.79	+1.25	17	HB	−2.40	−0.96	+1.44
May 19	FJ	+2.92	+1.76	−1.16	18	KB	−0.13	−0.59	−0.46
21	RL	+1.82	+3.29	+1.47	20	GW	−1.85	−0.22	+1.63
28	KB	+0.78	−0.63	−1.41	Sept. 14	FJ	+0.60	+0.64	+0.04
June 16	KB	−1.57	0.00	+1.57	16	KB	−2.17	−1.54	+0.63
18	FJ	+2.21	−0.46	−2.67	Oct. 13	FJ	+0.25	+0.89	+0.64
July 23	KB	+0.86	+0.18	−0.68	14	KB	−1.14	−2.16	−1.02
26	FJ	−2.39	−2.27	+0.12	24	FJ	+1.81	+0.45	−1.36
27	HB	−0.28	−0.87	−0.59	Dec. 23	FJ	−0.89	+0.85	+1.74
28	KB	−2.26	−2.32	−0.06					
Aug. 23	KB	−2.38	−1.28	+1.10					
Sept. 25	HB	−1.72	−2.11	−0.39					
Oct. 20	KB	−0.70	−0.34	+0.36					
23	KB	−1.78	−1.54	+0.24					
Dec. 15	KB	−1.93	−1.25	+0.68					
			Mean	+0.21				Mean	+0.86

Assuming that the effect of any error in the adopted Physical Libration of the Moon will disappear in the mean of these quantities, and allowing for the fact that the Tabular Diameter has been increased by +2.″40, the correction to the Moon's Tabular Vertical Diameter is

$$+2.''40 \quad +0.''21 \quad +0.''86 \quad = \quad +3.''47$$

This result is independent of any error in the assumed selenographical position of the crater. The observations indicate that the Tabular Declination of the crater, relatively to the Moon's centre, requires to be increased by

$$\tfrac{1}{2}\ (-0.''21 \quad +0.''86) \quad = \quad +0.''32$$

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MAJOR PLANETS

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
MERCURY											
1948 d h			h m s	s		° ′ ″	″	s	s	″	″
May 19 13	FJ	1	5 9 48.83	+.10	C	+25 16 16.61	+0.27	..	..	..	..
July 22 11	KB	2	6 45 2.51	+.12	C	+21 52 12.90	-0.32	..	..	..	..
23 11	GW	C	6 51 21.19	+.23	C	+21 58 56.10	-0.73	..	..	..	..
27 11	KB	C	7 19 38.03	+.07	C	+22 8 9.39	-0.72	..	..	..	..
Sept.28 13	KB	..	13 51 (15)	..	C	-14 44 44.29	-1.60	..	..	..	..

Mercury - The computed positions are for observed time of transit and have been interpolated from the 0^h ephemeris given in the Nautical Almanac.

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
VENUS											
Feb. 2 15	KB	1	23 21 2.37	+.06	S	- 5 24 11.89	+0.21	..	..	..	..
17 15	KB	1	0 26 8.53	+.20	S	+ 2 26 20.28	+0.78	..	..	..	..
26 15	RL	1	1 4 36.72	+.10	S	+ 7 5 54.36	-2.54	..	..	..	..
Mar. 1 15	HB	1	1 21 41.75	+.10	S	+ 9 6 43.12	+0.42	..	..	..	..
12 15	RL	1	2 8 55.06	+.08	S	+14 20 22.90	+0.30	..	..	..	..
18 15	KB	1	2 34 54.17	+.10	S	+16 56 8.02	-0.58	..	..	..	..
22 15	RL	1	2 52 18.68	+.05		+18 32 27.81	-0.09	..	..	20.52	+1.68
24 15	GW	1	3 1 2.31	+.18	S	+19 18 6.45	-1.35	..	..	..	..
Apr. 12 15	KB	1	4 23 43.29	+.11	S	+24 55 48.83	-0.87	..	..	..	..
15 15	HB	1	4 36 29.60	+.01	S	+25 31 40.15	-0.75	..	..	..	..
16 15	RL	1	4 40 42.72	-.03	S	+25 42 28.26	-2.64	..	..	..	..
22 15	KB	1	5 5 29.54	+.05	S	+26 35 43.92	-0.98	..	..	..	..
26 15	KB	1	5 21 20.93	+.02		+27 0 7.74	-1.16	..	..	27.88	+0.78
May 7 15	KB	1	6 0 53.31	+.05	S	+27 24 10.74	+0.74	..	..	..	..
18 15	RL	1	6 31 22.32	-.28		+26 53 27.67	+1.07	..	..	38.52	+1.42
19 15	FJ	1	6 33 31.83	-.20		+26 48 24.40	+0.20	..	..	40.39	+2.71
21 15	RL	1	6 37 28.96	-.08		+26 37 19.95	-1.15	..	..	40.25	+1.33
June 11 13	KB	1	6 42 30.94	-.06	S	+23 37 31.80	+1.40	..	..	..	..
July 6 11	FJ	2	5 45 17.31	-.09	N	+18 47 3.09	-0.71	..	..	..	..
22 10	KB	2	5 40 31.67	+.01		+17 50 52.44	+0.64	..	..	42.12	-0.90
27 9	KB	2	5 46 51.65	-.07		+17 58 13.66	+0.46	..	..	39.69	-0.03
30 9	FJ	2	5 52 8.26	+.11		+18 5 44.26	+0.06	..	..	38.51	+0.61

Mercury - May 19. Extremely ill-defined and unsteady.
July 22. Unsteady.
27. Poor object.
Sept.28. Very diffused and unsteady.

Venus - Feb. 2. Very diffused.
17. Unsteady.
Mar. 12. Unsteady.
24. Definition very bad.

Venus - Apr. 15. Very unsteady.
16. Very unsteady.
May 7. Very poor object.
18. Poor definition.
19. Very ill-defined and unsteady.
June 11. Extremely poor object.
July 6. Through cloud; very diffused and unsteady.
22. Unsteady; clouds passing.
30. Well-defined and steady.

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MAJOR PLANETS

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
VENUS											
1948 d h			h m s	s		° ′ ″	″	s	s	″	″
Aug. 18 9	KB	2	6 44 35.33	+.08		+18 55 47.96	−0.74	..	..	30.06	+1.24
26 9	KB	2	7 13 36.76	+.03		+18 55 1.92	−0.18	..	..	27.94	+1.88
Sept. 16 9	KB	2	8 40 1.56	+.03		+16 47 33.70	−2.20	..	..	21.85	+1.01
20 9	FJ	2	8 57 24.42	+.12		+15 59 56.80	−1.80	..	..	21.12	+1.02
22 9	KB	2	9 6 9.07	+.02		+15 33 19.30	−0.40	..	..	18.86	−0.88
25 9	KB	2	9 19 19.45	+.10		+14 49 50.08	−1.82	..	..	19.72	+0.50
Oct. 2 9	FJ	2	9 50 12.73	+.22		+12 52 47.91	−0.59	..	..	20.04	+1.90
5 9	FJ	2	10 3 28.58	+.15		+11 56 19.74	−0.36	..	..	19.71	+1.97
Nov. 25 10	KB	2	13 51 21.04	−.03	S	− 9 25 59.69	−1.99	..	..	..	..
26 10	GW	2	13 56 0.73	+.11	S	− 9 51 54.34	−1.54	..	..	..	..
MARS											
Jan. 29 2	KB		10 33 9.61	+.01		+13 42 9.96	−1.05	1.02	+.12	14.10	+1.12
Mar. 12 22	RL		9 34 18.99	−.09		+18 35 31.40	−0.60	0.96	+.08	14.92	+2.28
13 22	KB		9 33 26.53	−.03		+18 37 24.27	−0.03	0.91	+.03	14.83	+2.27
20 22	FJ		9 28 44.68	−.04		+18 42 36.96	+1.46	0.98	+.14	14.23	+2.29
22 21	RL		9 27 52.11	−.07		+18 41 38.94	−0.16	0.99	+.17	13.51	+1.75
23 21	KB		9 27 30.49	−.05		+18 40 47.19	−0.81	0.74	−.08	12.45	+0.77
Apr. 2 21	KB		9 26 38.39	−.08		+18 19 26.93	−1.67	0.79	+.03	12.45	+1.69
5 21	HB	1	9 27 18.22	−.07		+18 8 55.41	+0.41	..	..	12.12	+1.64
6 20	KB		9 27 36.83	−.08		+18 4 59.12	−1.08	0.71	−.01	11.46	+1.06
12 20	KB		9 30 21.72	−.12		+17 37 35.35	−1.75	0.70	.00	11.16	+1.28
13 20	GW		9 30 57.73	−.03		+17 32 26.11	+0.31	0.75	+.07	11.35	+1.55
15 20	HB		9 32 16.32	−.08		+17 21 32.72	+0.12	0.75	+.07	12.01	+2.37
19 20	FJ	1	9 35 19.29	−.11		+16 57 48.57	−0.43	..	..	10.83	+1.51
JUPITER											
May 25 2	FJ		17 45 22.08	−.12		−22 55 24.17	+0.63	3.30	+.02	43.15	+0.37
July 3 23	FJ		17 24 17.70	−.09		−22 47 43.79	+1.01	3.41	+.11	43.93	+1.23
29 21	RL		17 14 40.03	−.16		−22 42 52.07	−1.37	3.21	+.05	41.94	+1.24
30 21	FJ		17 14 26.85	−.07		−22 42 46.15	+0.65	3.29	+.15	42.99	+2.39
Aug. 14 20	KB		17 12 41.35	−.16		−22 43 25.59	+0.91	3.08	+.06	39.57	+0.59

Venus - Sept. 22. Diffused; clouds passing.

Mars - Mar. 12. Cloudy.
13. Very unsteady and diffused.
Apr. 2. Through clouds at times.

Mars - Apr. 6. Diffused.
19. Cloudy.

Jupiter - May 25. Observed through cloud.

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MAJOR PLANETS

Universal Time	Observer	Limb	Right Ascension Observed	Right Ascension O - C	Limb	Declination Observed	Declination O - C	Diameter Horizontal Observed	Diameter Horizontal O - C	Diameter Vertical Observed	Diameter Vertical O - C
SATURN											
1948 d h			h m s	s		° ′ ″	″	s	s	″	″
Mar. 12 22	RL		9 18 43.13	-.20		+16 59 24.15	-0.55	1.26	-.12	19.16	+1.32
13 22	KB		9 18 29.51	-.10		+17 0 27.94	+0.24	1.39	+.01	18.97	+1.15
20 21	FJ		9 17 2.80	-.09		+17 7 0.88	-0.22	1.46	+.08	19.79	+2.13
23 21	KB		9 16 30.89	-.12		+17 9 21.53	-1.47	1.33	-.05	18.86	+1.26
Apr. 2 21	KB		9 15 9.45	-.12		+17 15 13.40	-1.90	1.39	+.03	19.20	+1.88
5 20	HB		9 14 52.84	-.14		+17 16 23.64	+0.04	1.38	+.04	18.62	+1.38
12 20	KB		9 14 28.89	-.14		+17 17 52.29	-1.91	1.33	+.01	18.61	+1.57
13 20	GW		9 14 27.26	-.06		+17 17 58.49	-0.71	1.28	-.04	19.02	+2.02
18 19	GW		9 14 25.13	-.11		+17 17 53.05	-1.55	1.28	-.04	18.69	+1.83
URANUS											
Jan. 8 22	KB	C	5 30 18.64	-.05	C	+23 25 18.29	-0.31	..	..	..	..
Feb. 12 20	KB	C	5 26 8.37	-.02	C	+23 22 18.90	+0.40	..	..	..	..
16 20	RL	C	5 25 54.78	-.12	C	+23 22 6.62	-1.58	..	..	..	..
Sept. 26 6	KB	C	6 2 20.07	-.03	C	+23 37 50.61	-1.85	..	..	..	..
Oct. 23 4	KB	C	6 1 59.98	+.03	C	+23 38 23.78	-0.63	..	..	..	..
NEPTUNE											
May 2 22	KB	C	12 42 15.59	-.48	C	- 2 49 3.59	+1.61	..	..	..	..
6 22	RL	C	12 41 56.19	-.45	C	- 2 47 7.03	+0.67	..	..	..	..
7 22	KB	C	12 41 51.50	-.46	..	- 2 46 (40)	..	..	..	..	..
8 22	GW	C	12 41 46.96	-.40	C	- 2 46 10.84	+1.16	..	..	..	..
18 21	GW	C	12 41 5.42	-.46	C	- 2 42 6.34	+0.86	..	..	..	..
19 21	FJ	C	12 41 1.81	-.41	C	- 2 41 44.16	+1.94	..	..	..	..

Saturn - Mar. 23. Very diffused at times; unsteady.
Apr. 2. Through cloud at times.

Uranus - Jan. 8. Diffused.

Uranus - Feb. 12. Very faint and diffused.

Neptune - May 2. Very faint.

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